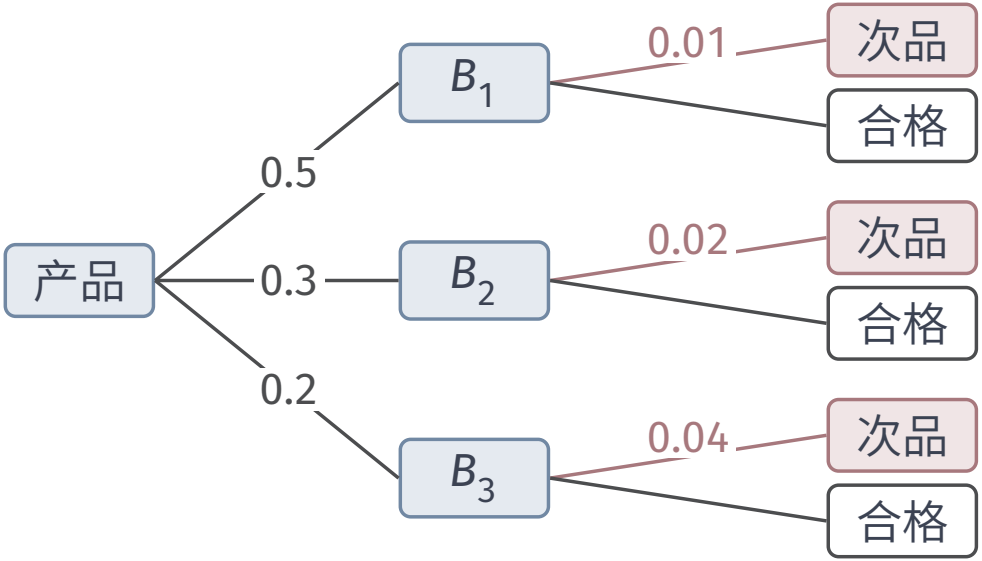

$0.5 \times 0.01 = 0.005$

$0.3 \times 0.02 = 0.006$

$0.2 \times 0.04 = 0.008$

$\mathbb{P}(D) = 0.019$

$$\mathbb{P}(B_3 \mid D) = \frac{0.008}{0.019}$$